RISHITA CHAWLA

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EDUCATION

Master of Science in Artificial Intelligence, Robotics Concentration *Expected May 2028*

Northeastern University - Recipient of International Impact Scholarship and Merit Scholarship *Boston,MA*

Coursework: Foundations of AI, Linear Algebra for Data Science

Bachelor of Science in Computer Engineering *August 2022 - May 2026*

University of North Carolina at Charlotte — Magna Cum Laude & Engineering Honors | GPA: 3.8 *Charlotte, NC*

Coursework: Embedded Systems, ML for IoT, Electronics, Signals & Systems, Data Structures & Algorithms, Computer Utilization in C++

TECHNICAL SKILLS

Languages: Python, C, C++, Embedded C, MATLAB

ML and Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn

Computer Vision & Robotics: OpenCV, YOLOv11, Instance Segmentation, ROS, Gazebo, RViz

MLOps: MLflow, FastAPI, Docker, REST APIs

Data & Developer Tools: NumPy, Pandas, Matplotlib, Seaborn, SQL, Git, GitHub, Linux

Embodied AI & Embedded Systems: NVIDIA Jetson, TensorRT, STM32, ESP32-S3-EYE, RTOS, GPIO, UART, I²C, SPI, CAN

WORK EXPERIENCE

Project Lead: Machine Learning Vision System for Axle Installation *August 2025 – May 2026*

Daimler Truck North America – Mt Holly, NC

- Led a team of six to develop an AI-powered computer vision system in Python for real-time axle alignment verification, achieving 0.25° angular and 5 mm lateral offset accuracy.
- Built an end-to-end vision pipeline using 10,000+ annotated images, training YOLOv11 instance segmentation models with data augmentation to achieve 98% axle and 95% chassis detection accuracy.
- Implemented OpenCV camera calibration and pixel-to-mm transformation for accurate offset measurement; deployed CUDA-accelerated inference on Nvidia Jetson achieving 40% faster inspections.

Robotics Research Intern – Autonomous Drone Swarm Target Search *March 2025 – December 2025*

SmartNet Labs – Charlotte, NC

- Developed a ROS-based coordination system for a three-drone tricopter swarm with TCP/IP communication and distributed task allocation, validated through autonomous search missions in Gazebo and Rviz simulation.
- Implemented Coverage Path Planning (CPP) and distributed area partitioning for multi-agent navigation, reducing target search time by 39%.
- Built an OpenCV perception and localization pipeline integrating GPS, IMU, and camera sensors, using object detection, coordinate transformations, and quaternion estimation, achieving 90% detection accuracy and 0.3 m localization error.

RELATED PROJECTS

Deep Learning Keyword Spotting System for Wake Word Detection

- Developed and benchmarked CNN, LSTM, and GRU models in TensorFlow/Keras for real-time keyword spotting, curating 2,000+ voice samples with pitch/time shifting and synthetic TTS augmentation, achieving 98% pre-quantization accuracy.
- Quantized the model to TensorFlow Lite, compressing model size from 650 KB to 90 KB, and deployed on real ESP32-S3-EYE embedded hardware using Embedded C, achieving 150 ms real-time inference latency.

STM32 RTOS-Based Virtual Pet with Custom SPI LCD Driver

- Developed a multithreaded virtual pet app on STM32F0 using CMSIS-RTOS2 (Render, Input, Logic threads) with mutex-protected LCD access and queue-based communication.
- Wrote a custom SPI1 RGB565 LCD driver from scratch, bringing up a 240x320 display and streaming animated frame data for real-time UI rendering.